

A Comparative Study of Classical and Transformer- Based Models for Phishing Email Detection

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Bachelor Semester Project 6

Why Phishing Email Detection?

Phishing emails are a common cybersecurity threat

They try to trick users into clicking links or revealing sensitive information

Automatic detection can help identify dangerous emails earlier

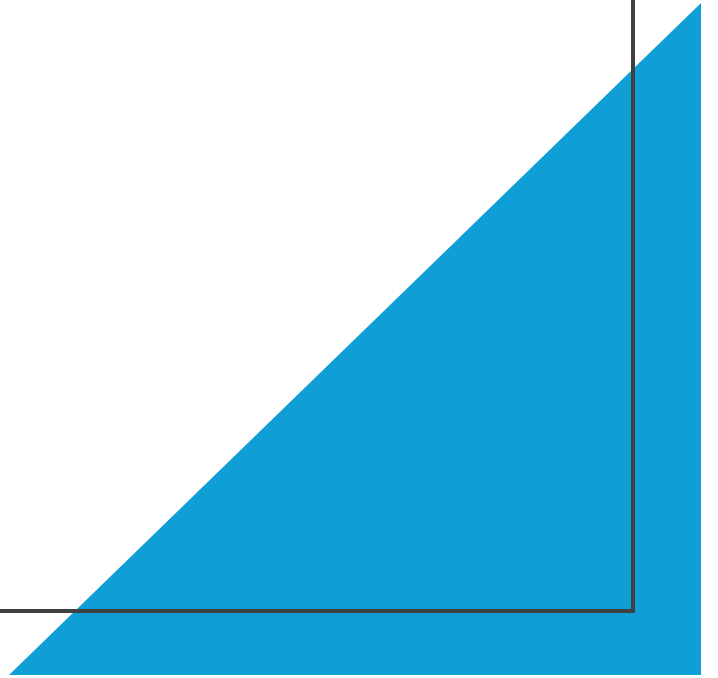
Research Objective

Scientific Question: How do classical feature-based models compare with transformer-based models for phishing email detection?

Classical model: TF-IDF + Logistic Regression

Transformer model: fine-tuned BERT

Focus: performance, interpretability, and generalization



Dataset

Primary dataset:

- zefang-liu/phishing-email-dataset
- 17,537 emails after preprocessing
- 10,979 safe / 6,558 phishing
- label = 0: Safe email
- label = 1: Phishing email

Second dataset:

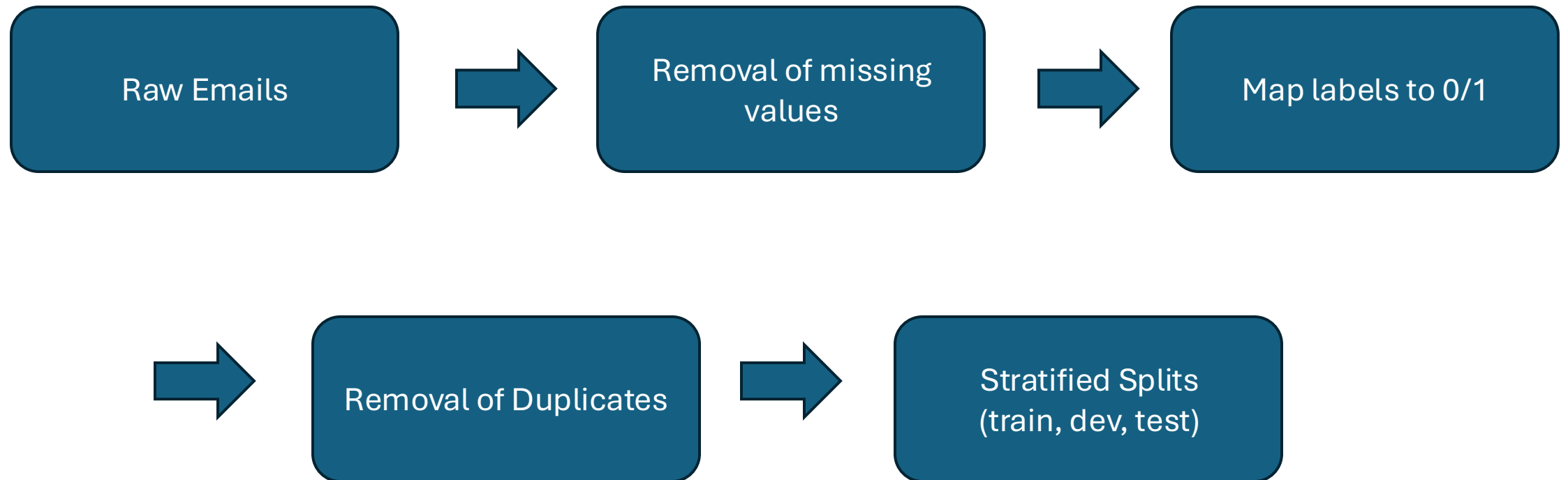
- DIFraud phishing benchmark
- Used for cross-dataset generalization



Dataset	Rows	Number of safe emails	Number of phishing emails	Phishing Percentage	Average Text Length
zefang-liu/phishing- email-dataset	17537	10979	6558	37.40	2861.99
difraud/difraud phishing	15272	9198	6074	39.77	2260.79



Preprocessing



Example Emails and Labels

	text	label
0	re : 6 . 1100 , disc : uniformitarianism , re ...	0
1	the other side of * galicismos * * galicismo *...	0
2	re : equistar deal tickets are you still avail...	0
4	software at incredibly low prices (86 % lower...	1

Models (Comparison)

TF-IDF + Logistic Regression

- Fast
- Simple
- Interpretable
- Coefficients can be inspected

BERT

- Contextual language model
- Understands word context
- Higher computational cost
- Less directly interpretable

Zero-shot NLI (Natural Language Inference)

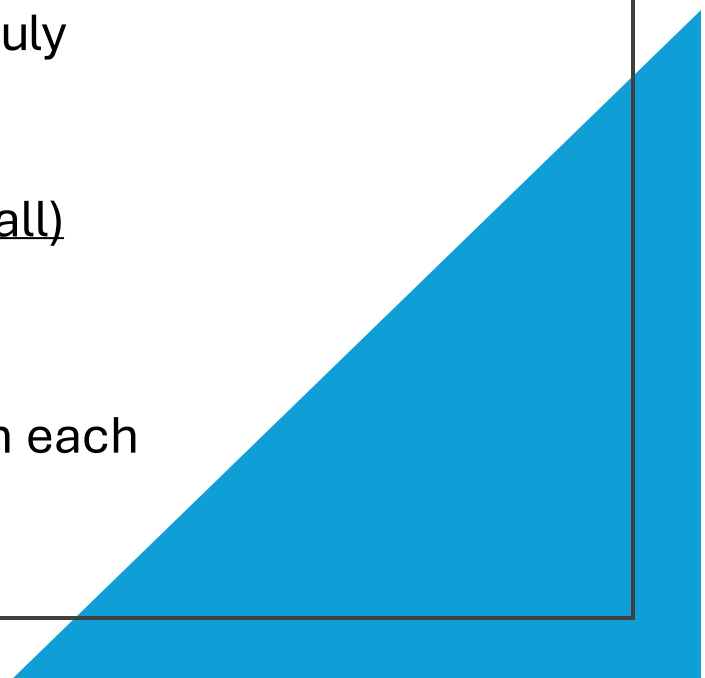
- No phishing-specific training
- Uses label hypotheses
- Evaluated on 200-email subset
- Exploratory baseline



Threshold Tuning

- **Default threshold:** 0.50
 - **Optimized threshold:** selected on the validation set for each model
 - **Goal:** maximize phishing F1-score
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Evaluation Metrics

- Accuracy (overall percentage of correct predictions)
 - Phishing precision (how many predicted phishing emails were truly phishing)
 - Phishing recall (how many real phishing emails were detected)
 - Phishing F1-score (balance between phishing precision and recall)
 - Macro-F1 (average F1 of both classes equally)
 - Micro-F1 (overall F1 across all predictions)
 - Weighted-F1 (average F1 weighted by the number of examples in each class)
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Main Results

MODEL	THRESHOLD	ACCURACY	PHISHING PRECISION	PHISHING RECALL	PHISHING F1-SCORE	MACRO F1- SCORE	MICRO F1- SCORE	WEIGHTED F1-SCORE
TF-IDF + Logistic Regression	0.45	0.9814	0.9708	0.9797	0.9752	0.9801	0.9814	0.9814
TF-IDF + Logistic Regression (balanced)	0.54	0.9814	0.9698	0.9807	0.9752	0.9802	0.9814	0.9814
BERT	0.08	0.9875	0.9847	0.9817	0.9832	0.9866	0.9875	0.9875
Zero-shot NLI	N/A	0.6200	0.4615	0.0800	0.1364	0.4464	0.6200	0.5239

Note: Zero-shot NLI was evaluated on a stratified subset of 200 test emails due to its higher computational cost.

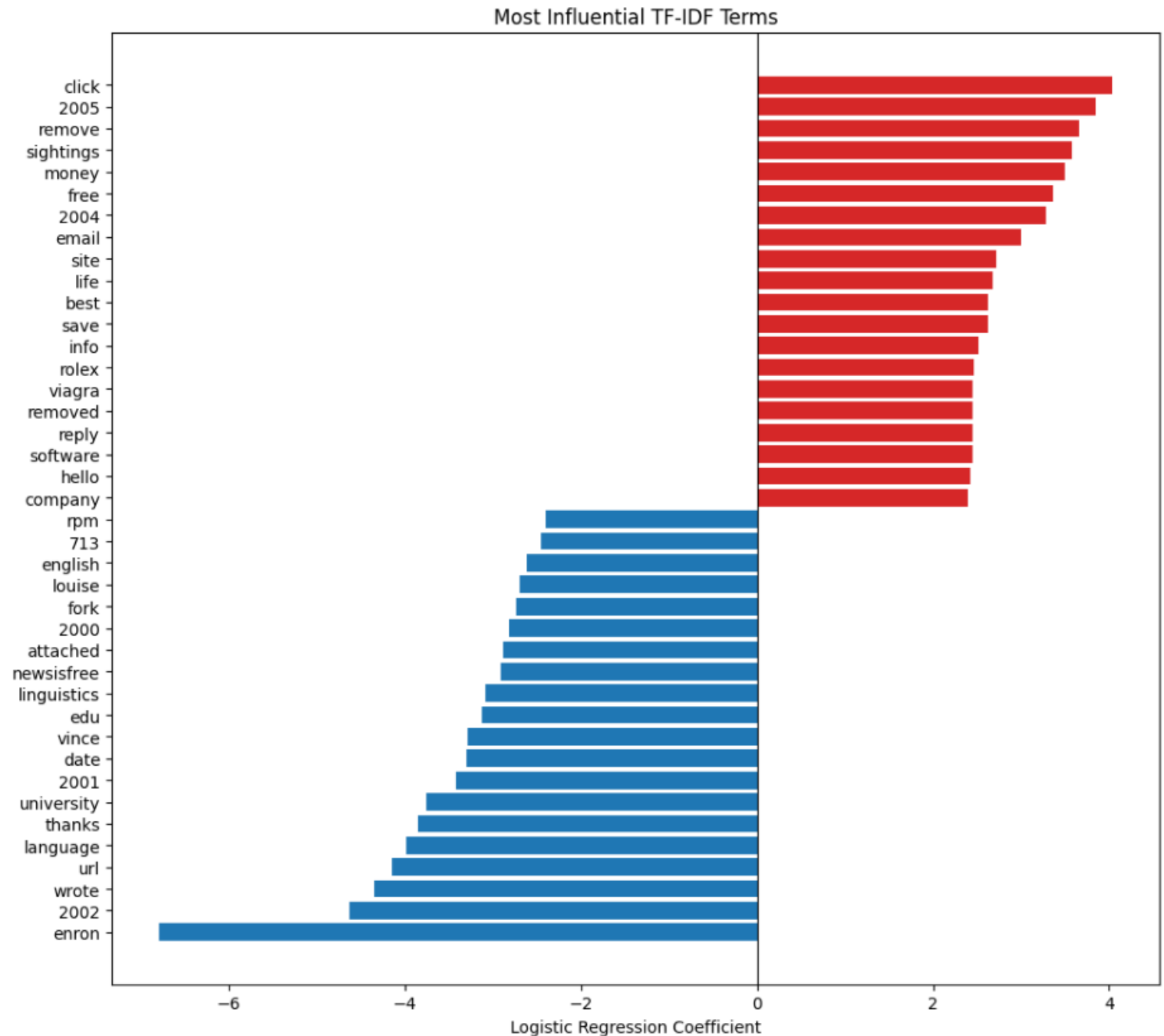


Explainability/Error Analysis

- Logistic Regression coefficients show which words push predictions toward phishing or safe
 - Some phishing emails look professional or technical
 - Some safe emails contain suspicious-looking links or formatting
 - This shows phishing detection is not only keyword matching
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Logistic Regression Explainability

- **Positive coefficients:** pushing the prediction toward the "phishing" category
- **Negative coefficients:** pushing the prediction toward the "safe" category
- Makes the classical model easier to interpret than BERT



Examples of Misclassified Emails

TF-IDF misclassified example

Text:

q : new email address for pier marco bertinetto content - length : 83 since the old address no longer works , would anybody know the new one ? alexis mr

True label: 0

Predicted label: 1

BERT misclassified example

Text:

Industrial LINUX News: The June issue of the ISA's InTech Magazine has an interesting article on how truly open Linux applications can lower development cost and increase f
This Linux news update brought to you by: www.Linux4oems.info -----If you don't want to receive fu
.naorwnwbxbsttgvelamusbs

True label: 1

Predicted label: 0

Cross-Dataset Generalization



Experiment	Accuracy	Phishing Precision	Phishing Recall	Phishing F1-score	Macro F1-score	Micro F1-score	Weighted F1-score
Primary train + test	0.9814	0.9708	0.9797	0.9752	0.9801	0.9814	0.9814
Difraud train + test	0.9719	0.9879	0.9408	0.9638	0.9704	0.9719	0.9717
Primary trained -> Difraud test	0.9051	0.8360	0.9474	0.8882	0.9029	0.9051	0.9059
Difraud trained -> Primary test	0.7948	0.9703	0.4654	0.6291	0.7436	0.7948	0.7725

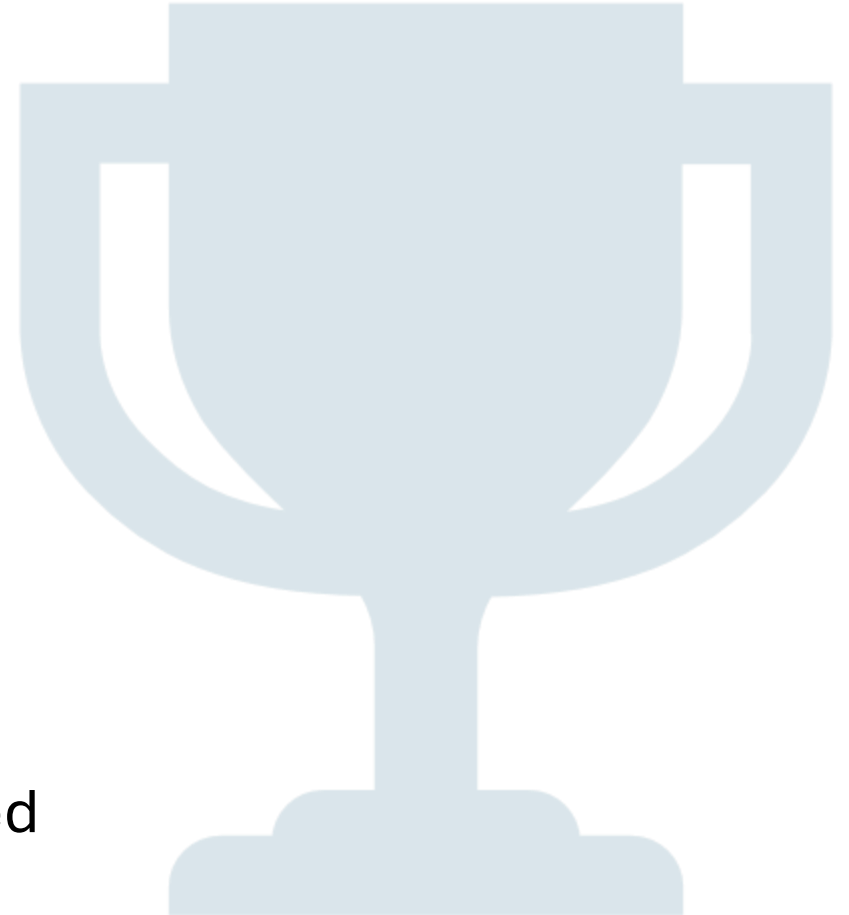
Limitations and Future Work

- BERT used max length 128 tokens
- Only one transformer model was fine-tuned
- Public phishing datasets may contain noisy labels
- Zero-shot NLI was evaluated on a 200-email stratified subset, but performed much worse than the trained models
- Test RoBERTa, DistilBERT, or long-context models
- Train on mixed datasets
- Add BERT explainability methods
- Evaluate deployment metrics such as latency and false negatives



Conclusion

- BERT achieved the best final performance
- TF-IDF + Logistic Regression remained highly competitive
- Zero-shot NLI was not competitive with the trained models
- Classical models are useful when speed and interpretability matter
- Cross-dataset testing showed that generalization remains a challenge



Thank you for
your attention!

